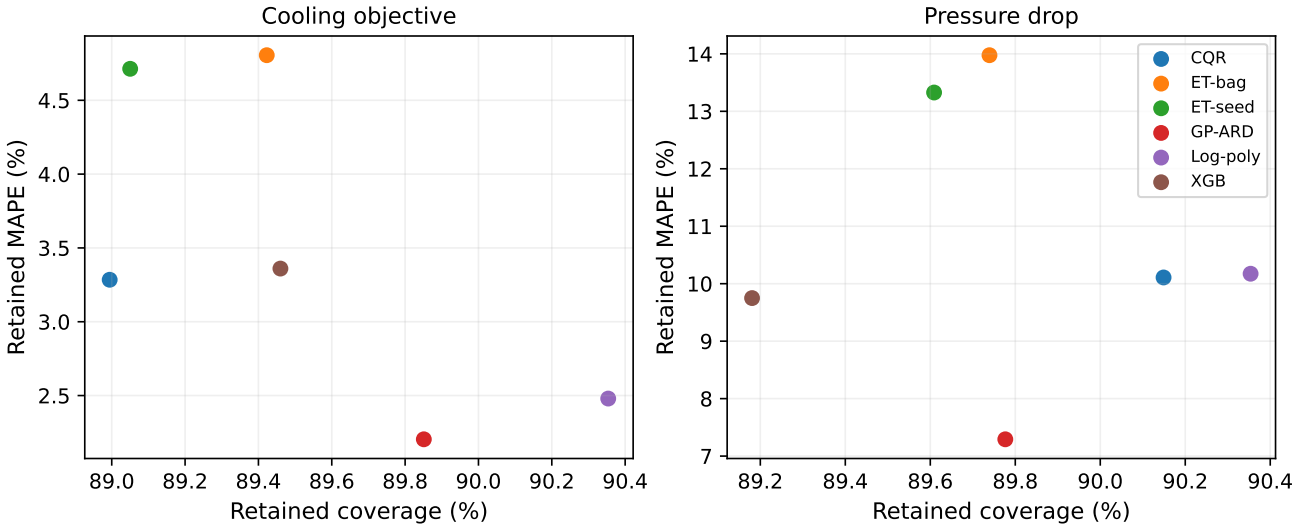
U-bend evaluation: figures

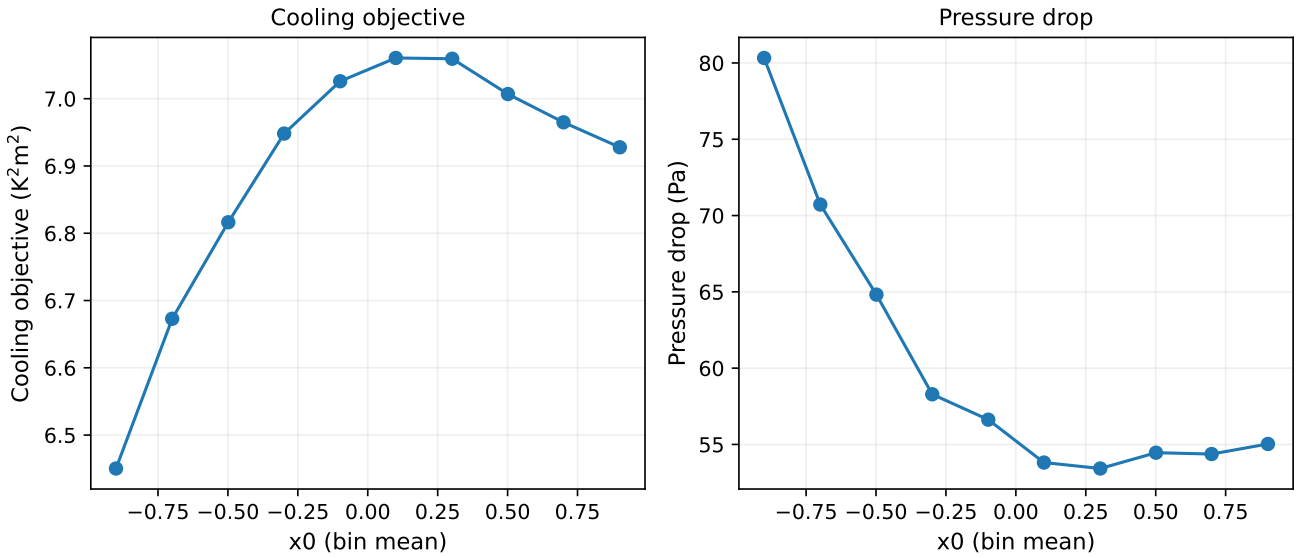
coverage

Design-random test: three-run mean retained MAPE versus measured coverage; the cutoff is calibrated independently at a nominal 90% budget. Models can return different populations.



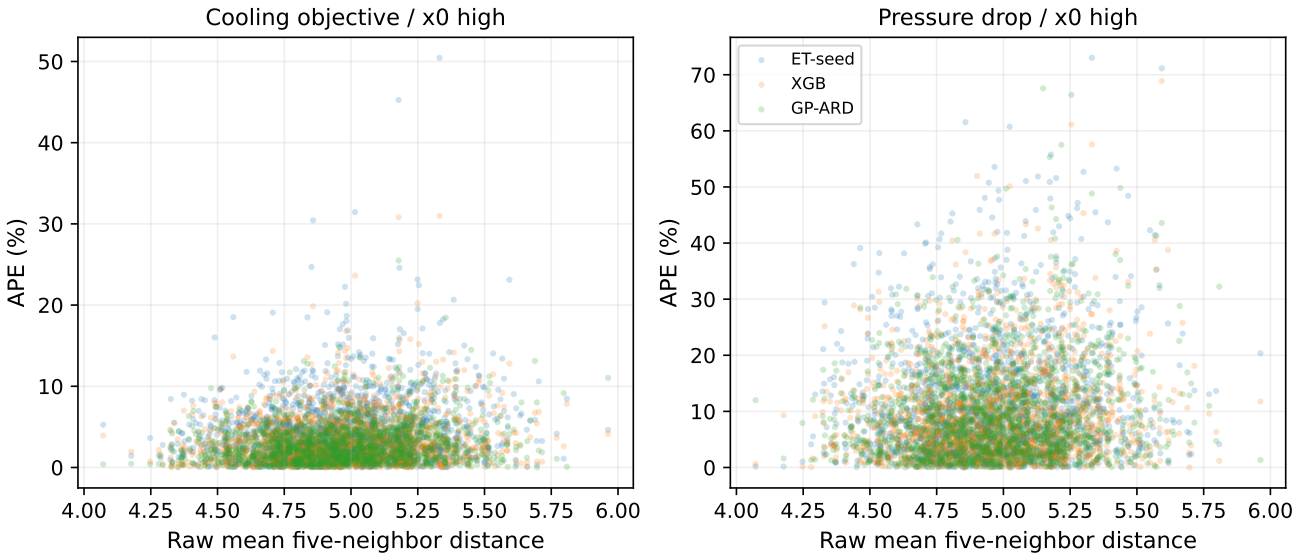
dataset

Response means in ten x0 bins across successful designs, averaged over other geometry coordinates; lines connect bin means, not additional CFD observations.



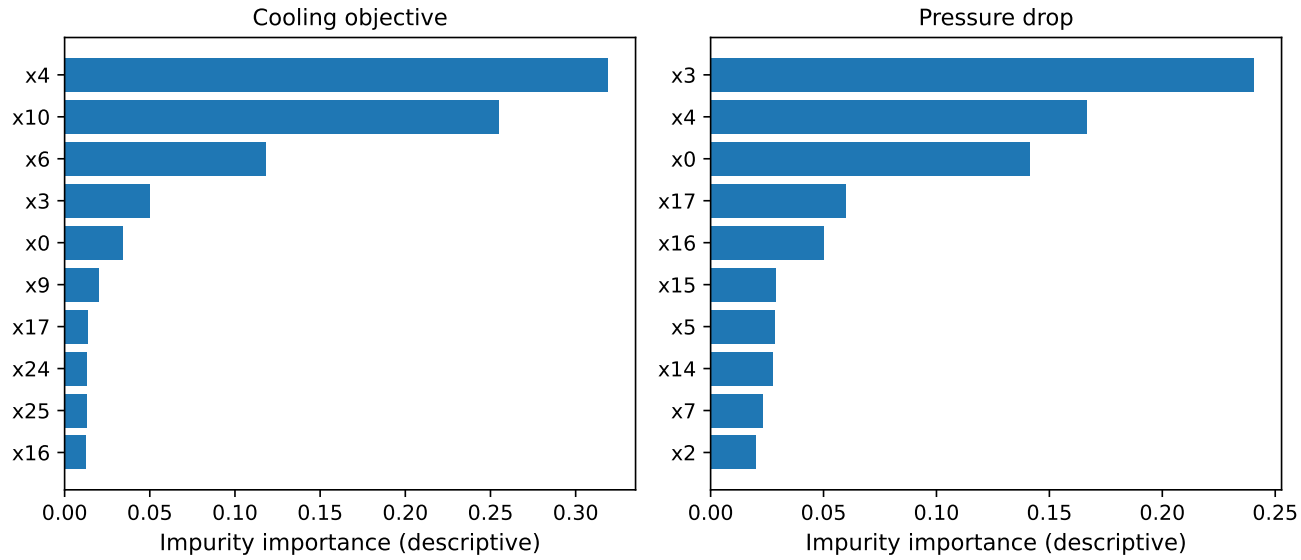
distance failure

Seed-42 high-x0 geometry-support test: response APE versus raw mean five-neighbor distance. Distances alone do not establish response error.



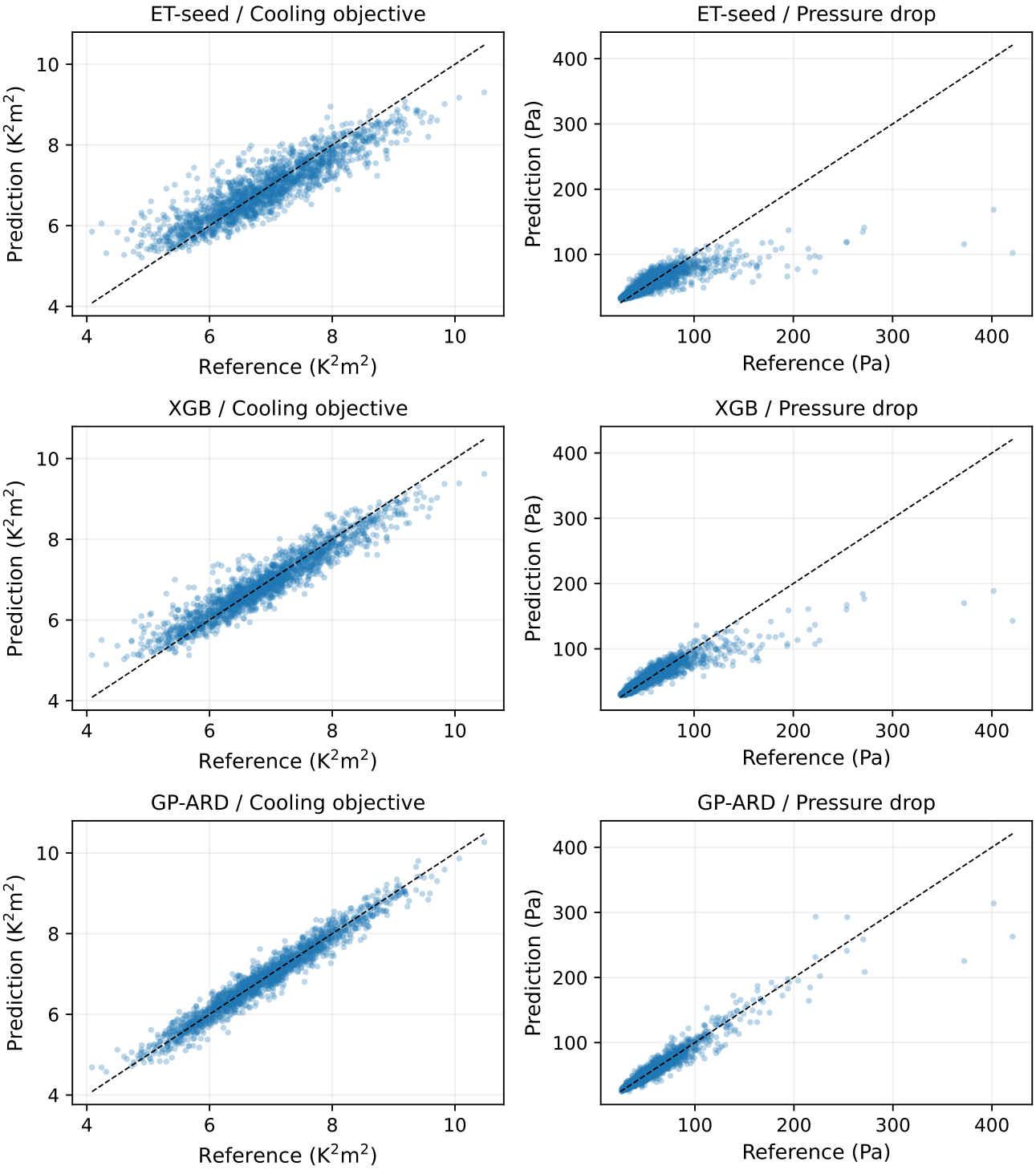
feature importance

Seed-42 ExtraTrees common-fit control on design-random data. Top ten impurity importances are descriptive, not causal physical effects.



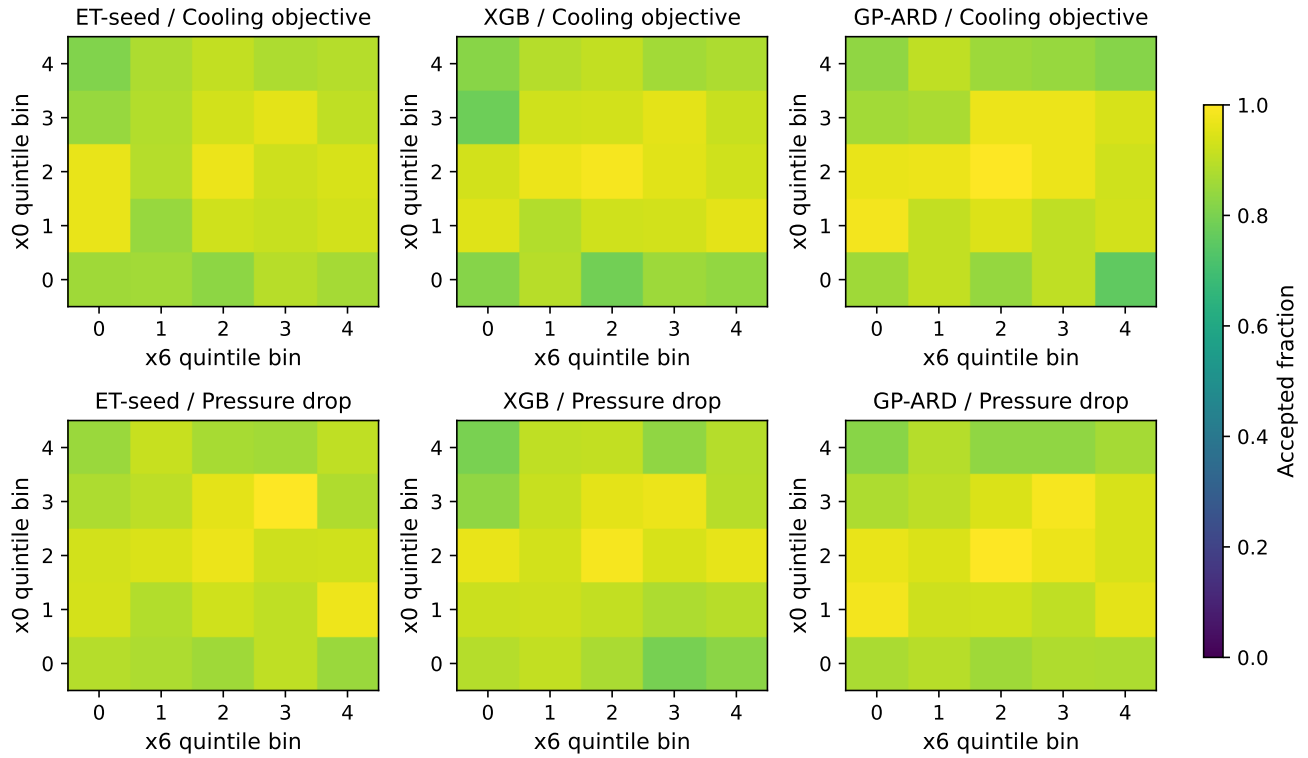
grouped parity

Seed-42 design-random test cases, same held-out designs across models. Dashed lines indicate exact agreement; axes show physical target units.



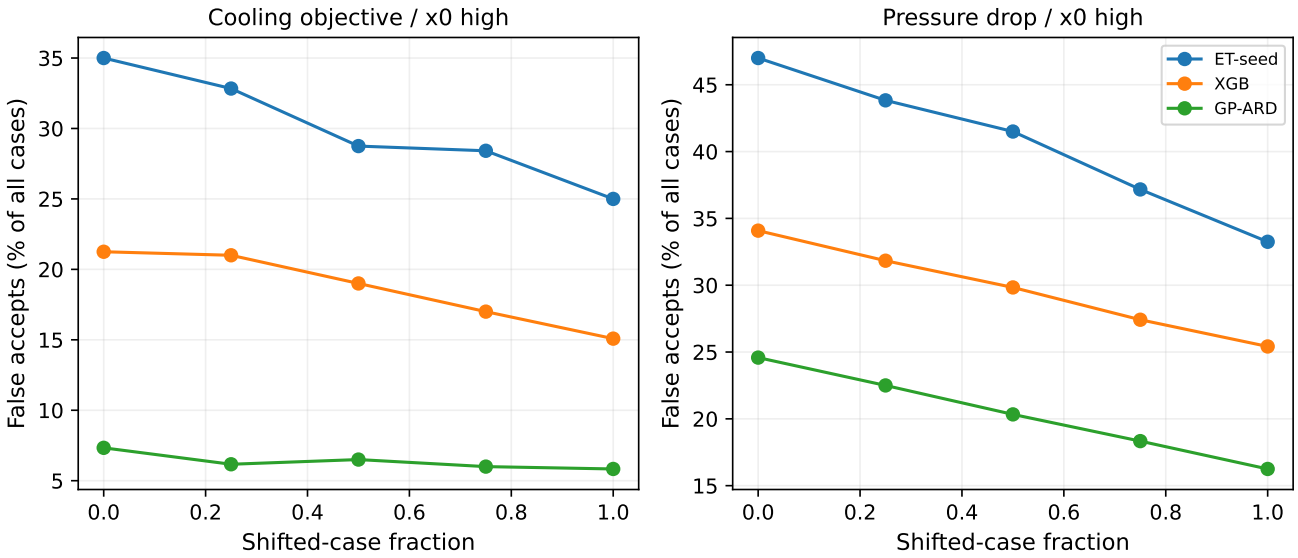
map

Seed-42 design-random accepted fractions in x0/x6 quintile cells, averaged over actual test cases and all remaining geometry coordinates. Cell counts are supplied; this is not a full-grid physical validity map.



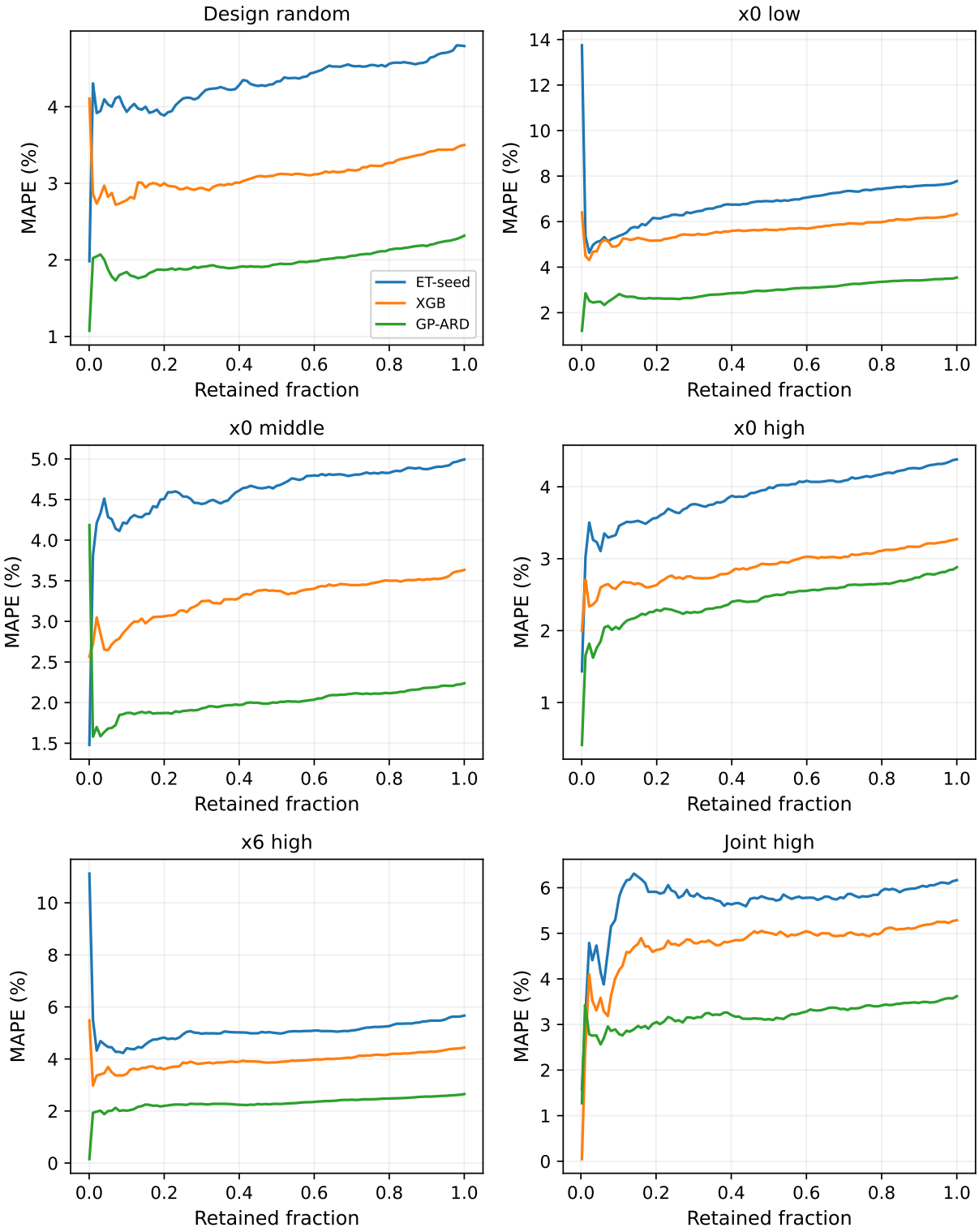
mixture

Common-model mixtures of supported and high-x0 test designs, averaged over three runs. False accepts use all test cases as the denominator; tolerances are illustrative.



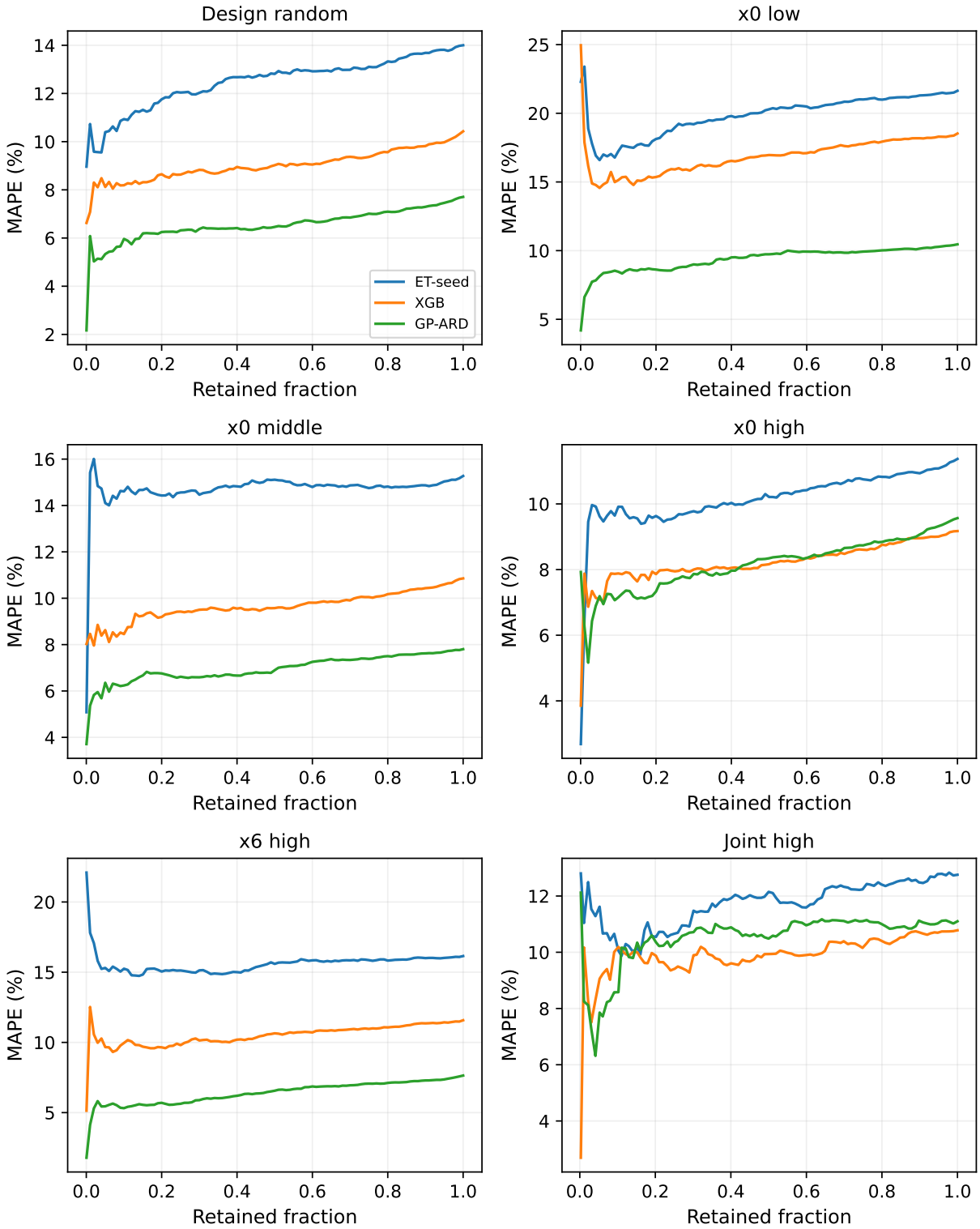
risk CoolingObjective K2m2

Cooling-objective seed-42 risk-coverage curves, separate folds, expected uniform ordering within score ties. These are retrospective rankings, not guaranteed deterministic threshold paths.



risk DelP Pa

Pressure-drop seed-42 risk-coverage curves under the same tie-aware, per-fold conventions; error is physical-unit MAPE in percent.



scores

Seed-42 design-random XGB uncertainty, distance and fused score distributions. The dashed line is the independently calibrated max-score cutoff.

